



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 4

Capacitors and RC Circuits

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Mid-term exam coming – Get ready!

- Keep in mind the **workload** (face-to-face & independent, self-directed study): at least **15 hours per week**.
- **Read carefully the announcement** about how the exam will work
 - Make sure you attempt the mock/practice exam.
- **Tips** to get ready for the exam:
 1. **Attend the workshops**
 - Essential to get familiarised with the analysis techniques and revise some "tricky" questions.
 - You can watch the recording if you have missed any.
 2. **Complete solved examples under Supplementary Material** (solutions provided)
 - **Essential** to fully understand the analysis techniques (if attempted by yourselves without checking the solution first).
 3. Complete **sample exams from previous years** (solutions provided)
 4. Complete the independent tutorials under Supplementary Material (video solutions provided) and exercises in the lecture slides.

Topic 4 Content

This lecture covers:

- Capacitors
- Circuit analysis with capacitors
- First order circuits with resistors and capacitors (RC circuits)
 - Natural response
 - Step response

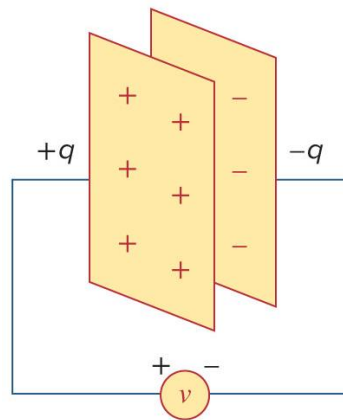
Corresponds to parts of Chapters 6 and 7 of your textbook

Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO2*: Obtain the transient and steady state behaviour of a first order circuit.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

Capacitors

- A capacitor is a circuit element that **stores energy** in its electric field.
 - It consists of two conducting plates separated by an insulator (or dielectric).
 - The plates are typically aluminum foil.
 - The dielectric is often air, ceramic, paper, plastic, or mica.
- When a voltage source v is connected to the capacitor, the source deposits a **positive charge** $+q$ on one plate and a **negative charge** $-q$ on the other plate.
 - The charges will be equal in magnitude on both plates.
 - The amount of charge is proportional to the voltage.



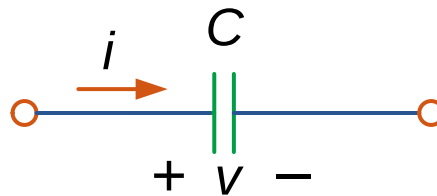
Capacitors

- The ratio of charge to voltage across a capacitor is its *capacitance*.

$$q = Cv$$

- The symbol for capacitance is C.
- Capacitance is measured in farads, F, which are coulombs per volt.

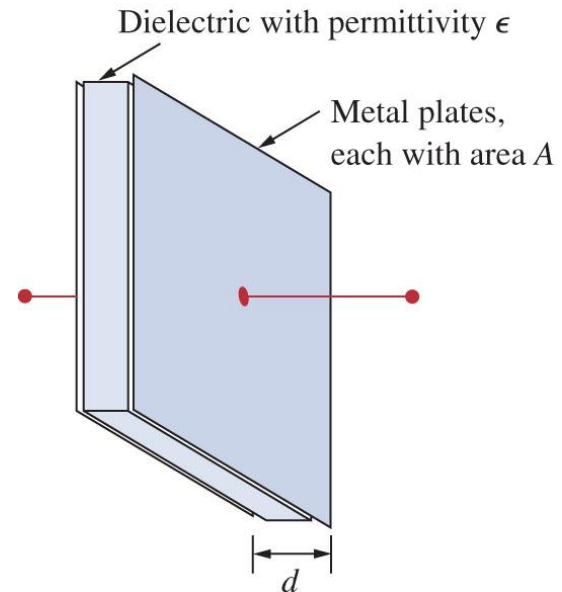
$$C = \frac{q}{v} = \frac{1 \text{ C}}{1 \text{ V}} = 1 \text{ F}$$



Capacitors

- Capacitance depends of the **physical dimensions** and **geometry** of the capacitor.
- For **parallel-plate** capacitors, the capacitance is given as follows:
 - A is the **surface** area of each plate.
 - d is the **distance** between the plates.
 - ϵ is the **permittivity** of the dielectric.

$$C = \frac{\epsilon A}{d}$$



Capacitors

- Capacitors are available in different values and types.
- They are described by their **dielectric material**.
 - E.g. thin film capacitors with polyester (Fig. 1(a)), ceramic (Fig. 1(b)), or electrolytic (Fig. 1(c)).
- Most capacitors are rated in picofarad (pF) to microfarad (μF).
- They are used to block DC and pass AC signals, shift phase, store energy, suppress and filter noise, etc.

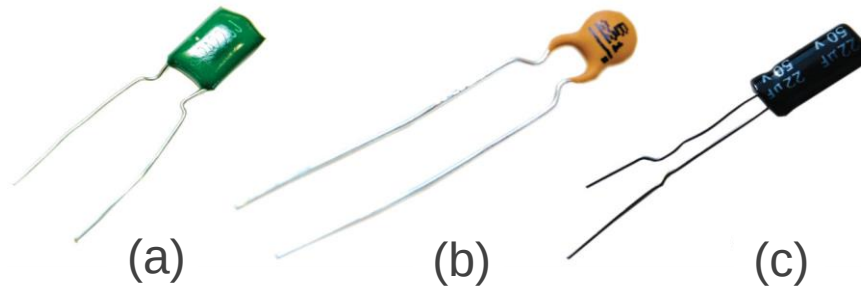
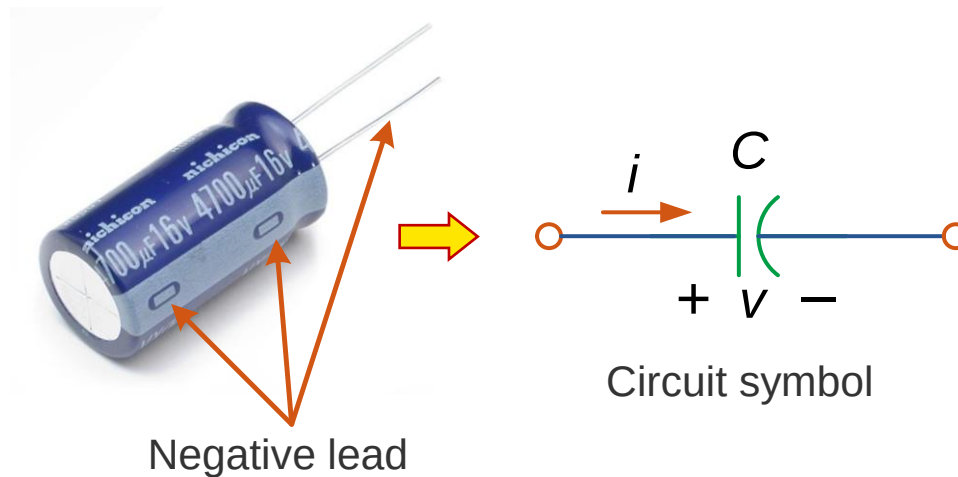


Fig. 1

Capacitors

- **Passive sign convention** applies to capacitors as well.
 - If $v \times i > 0$, the capacitor is being **charged** (absorbing energy).
 - If $v \times i < 0$, the capacitor is **discharging** (supplying energy).
- For capacitors in the range of μF , particularly **electrolytic** ones, the **polarity** is **already assigned** (the **negative** sign is marked on the capacitor).



Capacitors

- Charge = Capacitance x Voltage

$$q = Cv$$

- Differentiation with respect to time gives:

$$\frac{dq}{dt} = C \frac{dv}{dt}$$

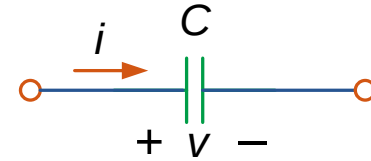
- Current is rate of change of charge, so

$$i = C \frac{dv}{dt}$$

- Voltage will then be:

$$v(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$$

where $v(t_0) = \frac{q(t_0)}{C}$ is called the **initial voltage** or **initial conditions** at time t_0 .



Capacitor's current i is proportional to the **rate of change** of its **voltage v** , with **capacitance C** as the constant of proportionality, assuming **passive sign convention**.

Capacitors

- **Instantaneous power** delivered to the capacitor:

$$p = vi = Cv \frac{dv}{dt}$$

- The **energy** stored in the **electric field** that exists between the plates of the capacitor can be then calculated as:

$$w(t) = \int_{t_0}^t p(\tau) d\tau = \int_{t_0}^t Cv(\tau) \frac{dv(\tau)}{d\tau} d\tau = C \int_{t_0}^t v(\tau) dv(\tau) = C \frac{v(\tau)^2}{2} \Big|_{t_0}^t$$

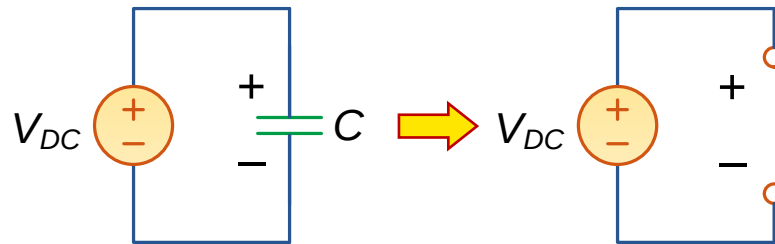
$$w(t) = \frac{1}{2} C (v(t)^2 - v(t_0)^2)$$

$$\text{If } v(t_0) = 0 \rightarrow w(t) = \frac{1}{2} C v(t)^2$$

Properties of capacitors

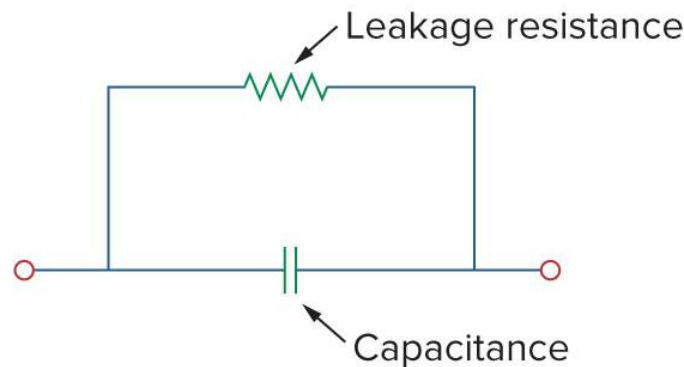
- A capacitor is an **open circuit** to **DC voltage**.

$$i = C \frac{dv}{dt} = 0 \text{ for constant voltage.}$$



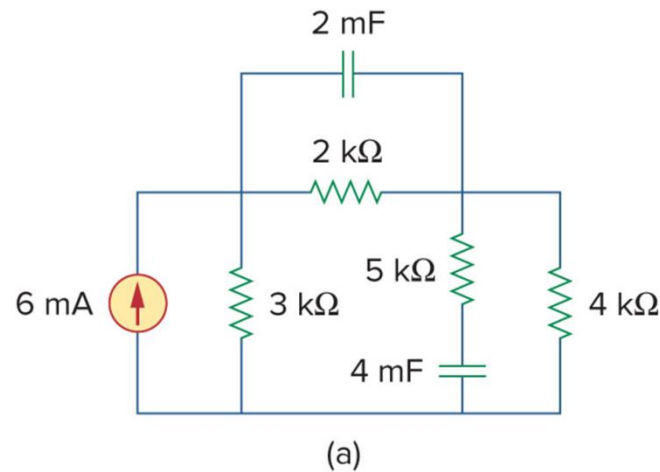
Properties of capacitors

- An **ideal capacitor** does not dissipate energy.
 - Energy is absorbed from a circuit, stored as electric field and then released back to the circuit.
- A **real capacitor** has a parallel-model **leakage resistance**, leading to a slow loss of the stored energy internally.



Exercise 1

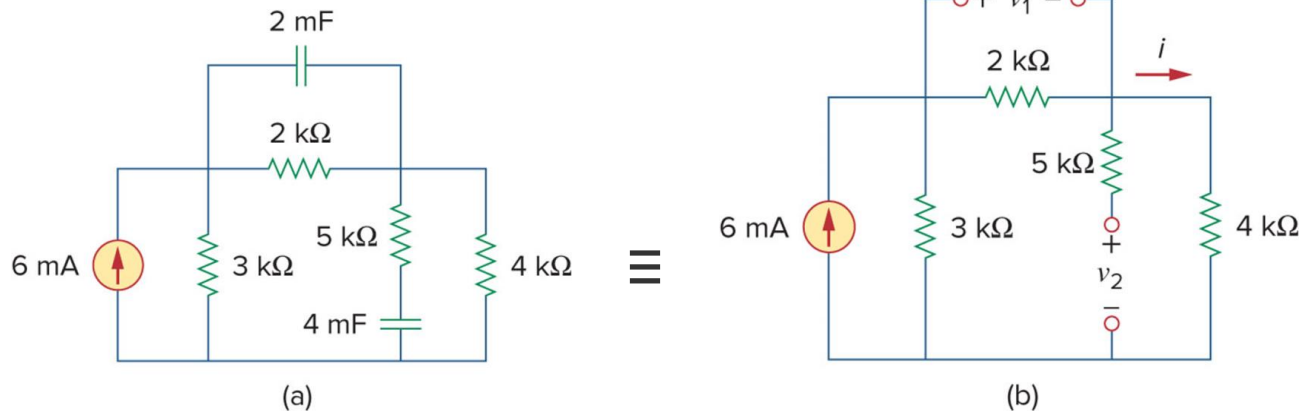
Obtain the energy stored in each capacitor in the circuit below (keep in mind they are under DC conditions)



$w(t) = \frac{1}{2} C v(t)^2$. Also, $i = C \frac{dv}{dt} = 0$ for DC voltage/current.

Exercise 1

Obtain the energy stored in each capacitor in the circuit below (keep in mind they are under DC conditions)



- To find the energy, we need the voltage across each capacitor. The easiest way is to first find the current i (current divider).

$$i = \frac{3 \parallel (2 + 4)}{(2 + 4)} \times 6 = \frac{1}{2 + 4} \frac{3 \times (2 + 4)}{3 + 2 + 4} \times 6 = \frac{3}{3 + 2 + 4} \times 6 = 2 \text{ mA} \rightarrow \begin{aligned} v_1 &= 2 \text{ k}\Omega \times 2 \text{ mA} = 4 \text{ V} \\ v_2 &= 4 \text{ k}\Omega \times 2 \text{ mA} = 8 \text{ V} \end{aligned}$$

$$w_1 = \frac{1}{2} C_1 v_1^2 = \frac{1}{2} (2 \times 10^{-3}) \times 4^2 = 16 \text{ mJ}$$

$$w_2 = \frac{1}{2} C_2 v_2^2 = \frac{1}{2} (4 \times 10^{-3}) \times 8^2 = 128 \text{ mJ}$$

Capacitors in parallel

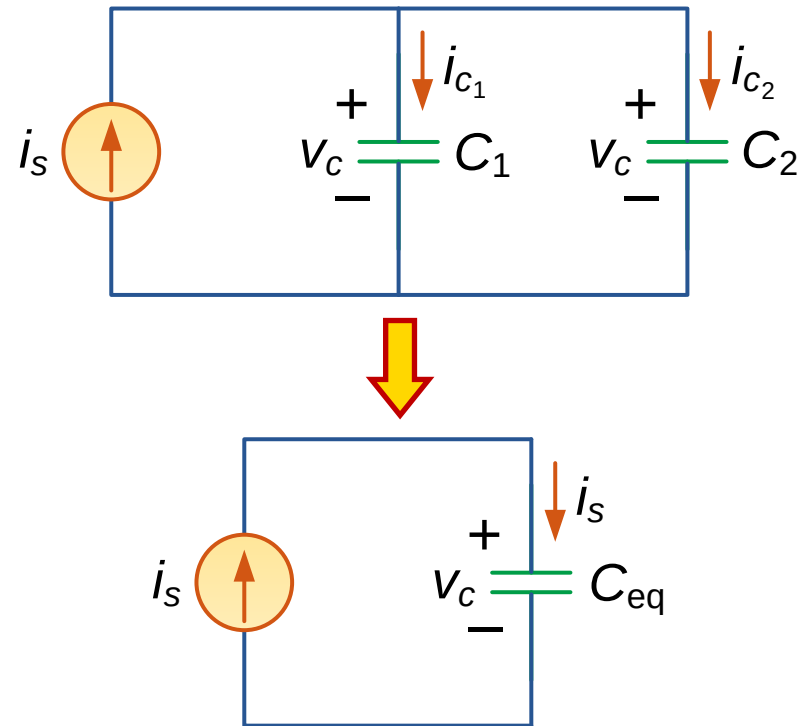
- Similar to resistors, capacitors in series or parallel can be combined to simplify the circuit.
- For capacitors in parallel, the voltage is the same across each capacitor.
- Applying KCL and current-voltage relation $i = C \frac{dv}{dt}$ for capacitors:

$$i_s = i_{c_1} + i_{c_2}$$

$$i_s = C_1 \frac{dv_c}{dt} + C_2 \frac{dv_c}{dt}$$

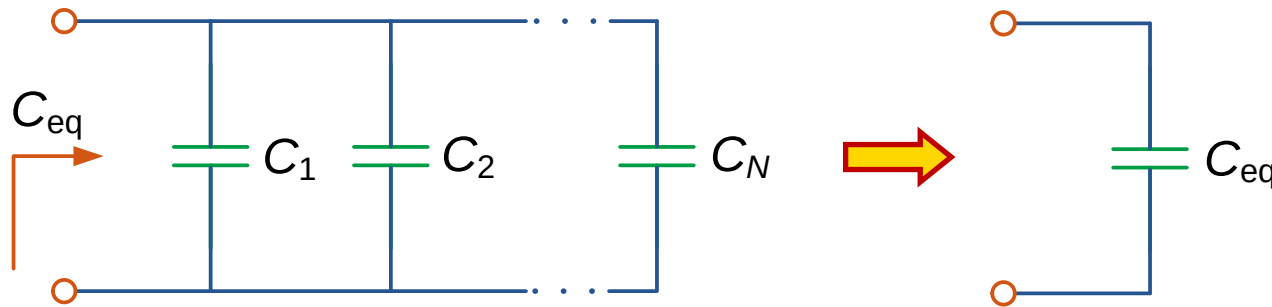
$$i_s = (C_1 + C_2) \frac{dv_c}{dt} \quad \leftrightarrow \quad i_s = C_{eq} \frac{dv_c}{dt}$$

$$C_{eq} = C_1 + C_2$$



Capacitors in parallel

- The **equivalent capacitance** of any number of **capacitors in parallel** is the **sum of the individual capacitances**.



$$C_{eq} = C_1 + C_2 + \cdots + C_N = \sum_{k=1}^N C_k$$

Capacitors in series

- For capacitors in series, the current is the same through each capacitor.
- Applying KVL and voltage-current relation for capacitors:

$$v_s = v_{c_1} + v_{c_2}$$

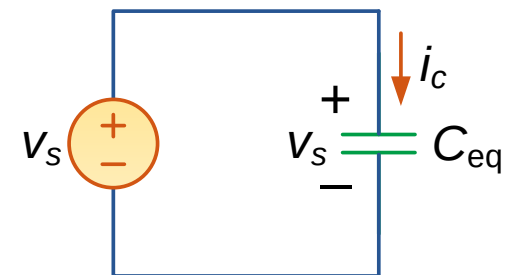
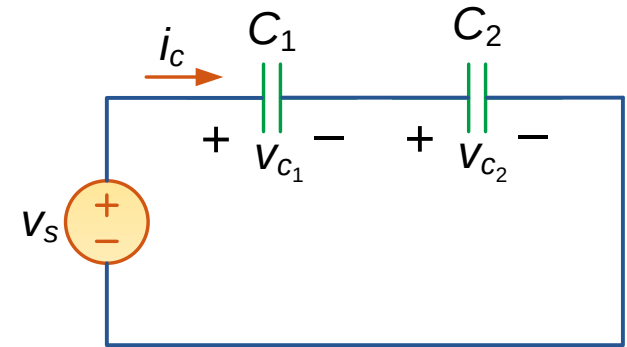
$$v_s = \frac{1}{C_1} \int_{t_0}^t i_c(\tau) d\tau + v_{c_1}(t_0) + \frac{1}{C_2} \int_{t_0}^t i_c(\tau) d\tau + v_{c_2}(t_0)$$

$$v_s = \left(\frac{1}{C_1} + \frac{1}{C_2} \right) \int_{t_0}^t i_c(\tau) d\tau + (v_{c_1}(t_0) + v_{c_2}(t_0))$$

$$\Leftrightarrow v_s = \frac{1}{C_{eq}} \int_{t_0}^t i_c(\tau) d\tau + v_{c_{eq}}(t_0)$$

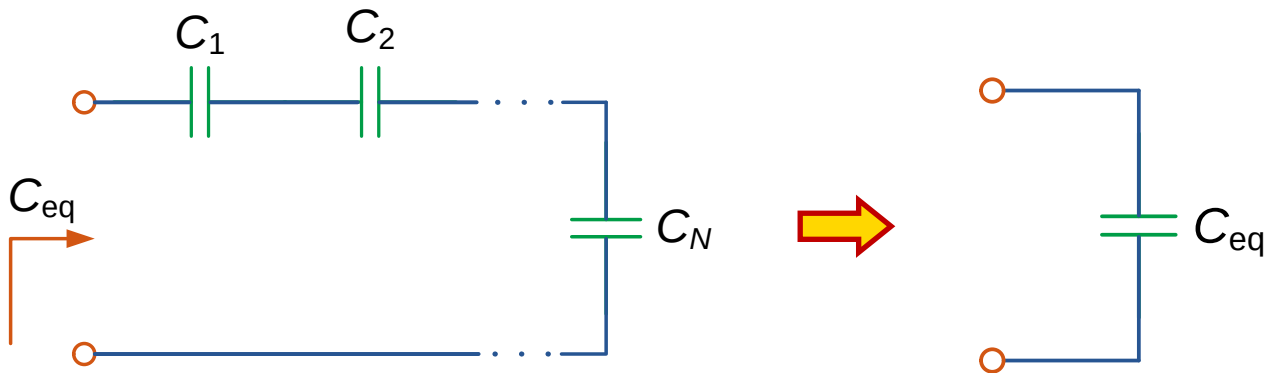
$$\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}} \quad \text{or} \quad \boxed{C_{eq} = \frac{C_1 C_2}{C_1 + C_2}}$$

$$\text{and } \boxed{v_{c_{eq}}(t_0) = v_{c_1}(t_0) + v_{c_2}(t_0)}$$



Capacitors in series

- The **reciprocal** of the **equivalent capacitance** of any number of **capacitors in series** is the **sum** of the **individual reciprocal capacitances**.



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \cdots + \frac{1}{C_N}$$

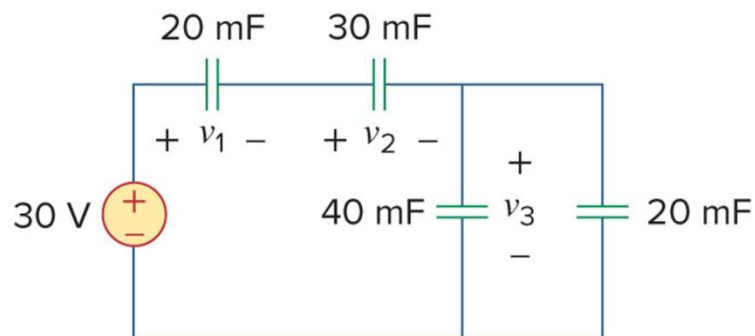
or

$$C_{eq} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} \cdots + \frac{1}{C_N}}$$

NOTE: Although the voltage across each capacitor is different for different values of capacitance, the **charge** across the plates is **equal** because charge is the integral of current, and current is the same through all of them.

Exercise 2

Find the voltage across and the charge on each capacitor in the given circuit.



Exercise 2

Find the voltage across and the charge on each capacitor in the given circuit

- First find the equivalent capacitance:

$$\frac{1}{C_{eq}} = \frac{1}{40 + 20} + \frac{1}{20} + \frac{1}{30} = \frac{1}{10} \rightarrow C_{eq} = 10 \text{ mF}$$

- Then, use $q = Cv$ to find total charge on equivalent capacitor:

$$q = C_{eq}v = 10 \times 10^{-3} \times 30 = 0.3 \text{ C}$$

- Charge is the same for capacitors in series, so:

$$q_{1-20\text{mF}} = q_{2-30\text{mF}} = 0.3 \text{ C}$$

- Therefore:

$$v_1 = \frac{q_1}{C_1} = \frac{0.3}{20 \times 10^{-3}} = 15 \text{ V} ; v_2 = \frac{q_2}{C_2} = \frac{0.3}{30 \times 10^{-3}} = 10 \text{ V}$$

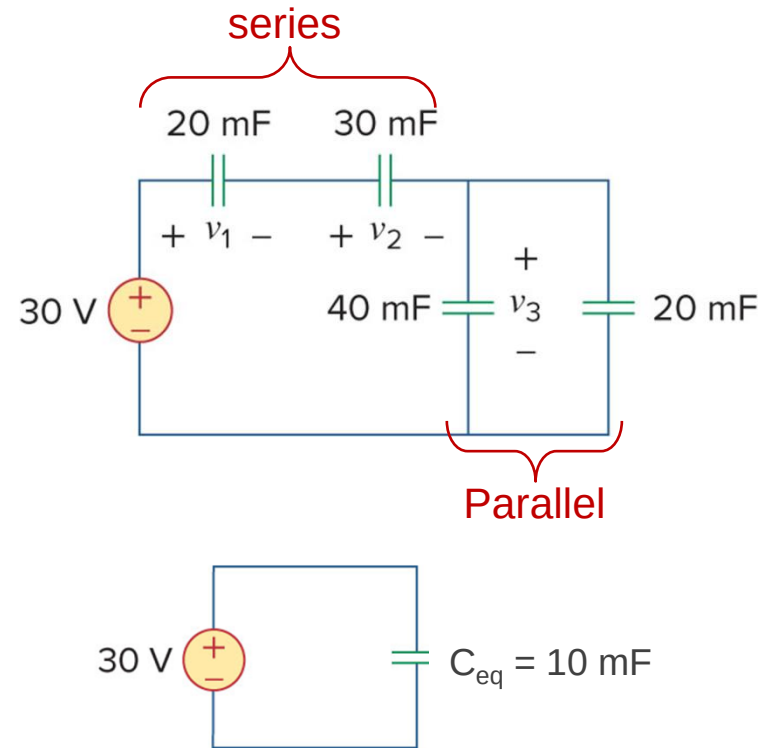
- Use KVL:

$$v_3 = 30 - v_1 - v_2 = 5 \text{ V}$$

- Use $q = Cv$ for 40-mF and 20-mF as they are in parallel:

$$q_{3-20\text{mF}} = 20 \times 10^{-3} \times 5 = 0.1 \text{ C}$$

$$q_{4-40\text{mF}} = 40 \times 10^{-3} \times 5 = 0.2 \text{ C}$$

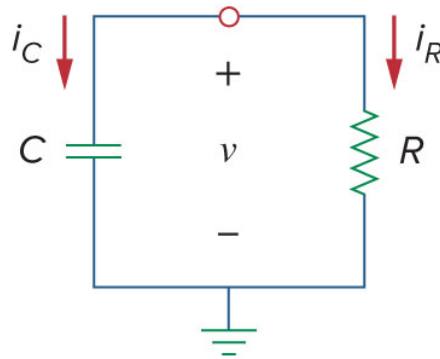


Circuit response

- The **response** of a circuit is the way it reacts to an excitation.
 - **Natural response** → Behaviour of the circuit (in terms of voltage or current) due to **initial energy stored** and **physical characteristics**.
 - **Forced response** → Behaviour of the circuit (in terms of voltage or current) due to **external sources** and **excitation**.

Natural response of RC circuit

Let's consider a circuit with a single capacitor, charged to an initial voltage V_0 and connected to a resistor. How will it behave?



We can use KCL to get:

$$-i_C - i_R = 0 \rightarrow -i_C = i_R$$

$$-C \frac{dv}{dt} = \frac{v}{R}$$

Note: We will consider how this capacitor was charged later in this lecture.

Note: A circuit characterised by a first order differential equation is called a first order circuit.

Natural response of RC circuit

Solve differential equation:

Rearrange:

$$-C \frac{dv}{dt} = \frac{v}{R}$$

$$\frac{dv}{dt} = \frac{-v}{RC}$$

Separate:

$$\frac{1}{v} dv = \frac{-1}{RC} dt$$

Integrate:

$$\int \frac{1}{v} dv = \frac{-1}{RC} \int dt$$

$$\ln(v) = \frac{-t}{RC} + D$$

Solve for $v(t)$

$$v(t) = e^{\frac{-t}{RC} + D}$$

$$v(t) = e^{\frac{-t}{RC}} e^D$$

$$v(t) = A e^{\frac{-t}{RC}}$$

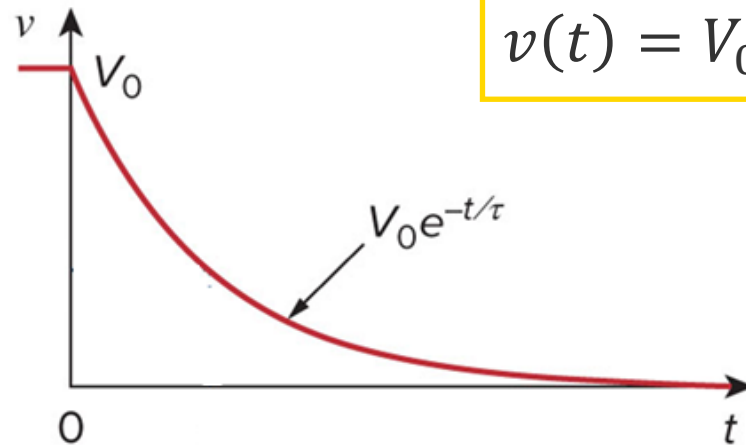
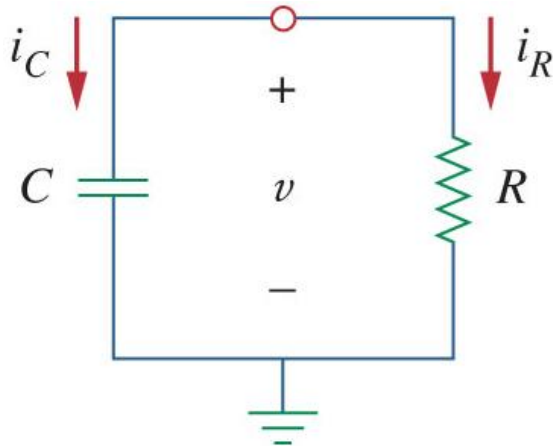
Apply initial conditions

$$v(0) = A e^0 = A = V_0$$

$$v(t) = V_0 e^{\frac{-t}{RC}}$$

Natural response of RC circuit

- There is no need to derive the differential equation solution every time, just use the result.
- The result shows that the **voltage response** of the RC circuit is an **exponential decay** of the **initial voltage**.



$$v(t) = V_0 e^{-\frac{t}{RC}}$$

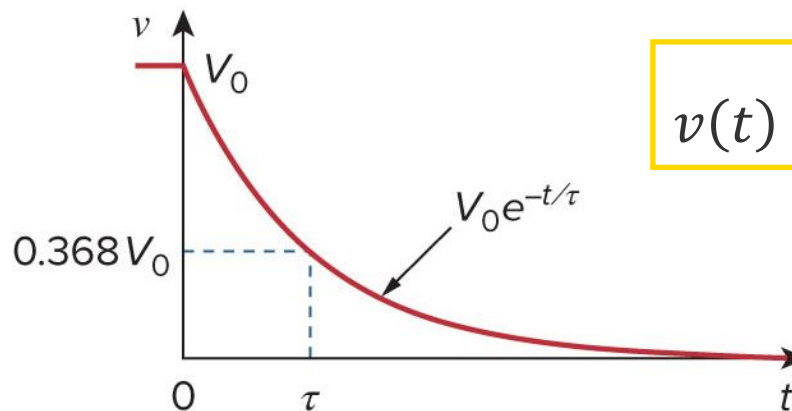
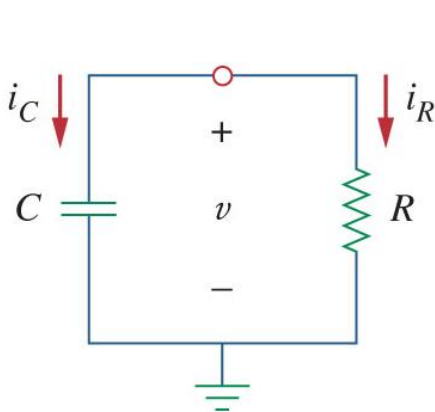
Time constant

- The **speed** at which the voltage **decays** depends on the **coefficient** of t in the power of exponential function, which can be expressed in terms of the *time constant*.

The **time constant** of a circuit is the time required for the response to **decay** to $1/e$ (or **36.8%**) of its initial value or to **increase** to $1 - 1/e$ (or **63.2%**) of its final value. It is denoted by τ .

- This implies that at $t = \tau$, the voltage should be $0.368V_0$:

$$V_0 e^{-\frac{t}{RC}} = V_0 e^{-1} = 0.368V_0 \rightarrow \tau = RC$$



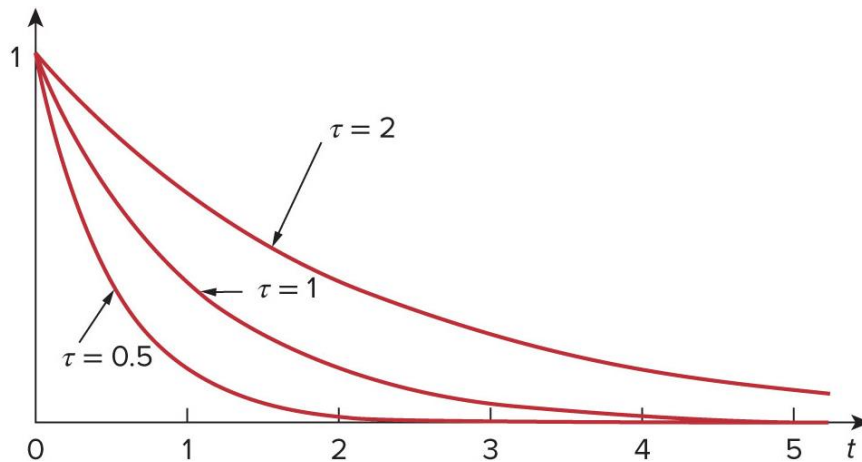
$$v(t) = V_0 e^{-\frac{t}{RC}} = V_0 e^{-\frac{t}{\tau}}$$

$$\tau = RC$$

Time constant

- For every time interval of τ , the voltage is reduced to 36.8 percent of its previous value.
 - A circuit with a **small time constant** has a **fast response** and **vice versa**.
- After **5 time constants**, the voltage $v(t)$ on the capacitor is less than **one** percent of its initial value V_0 .

It takes 5τ for an RC circuit to reach its **final state** or **steady-state** (either fully charged or fully discharged).



t	$\frac{v(t)}{V_0} = e^{\frac{-t}{\tau}}$
τ	0.36788
2τ	0.13534
3τ	0.04974
4τ	0.01832
5τ	0.00674

Natural response of RC circuit

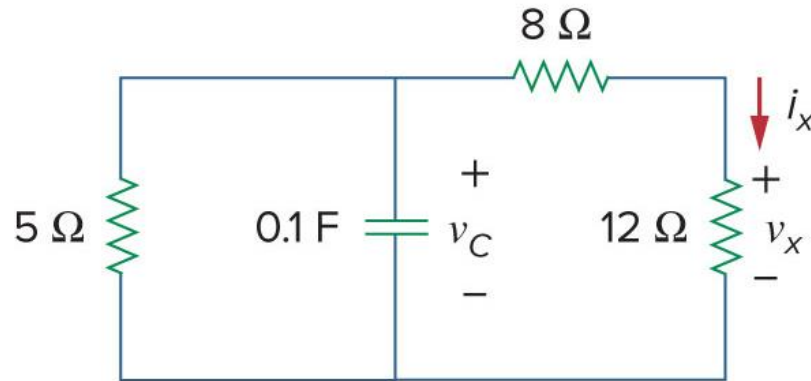
- Follow these steps to find the natural (or source-free response) of RC circuits:
 1. Find the **initial voltage** $v(0) = V_0$ across the capacitor **before** it is connected to the resistor.
 - The capacitor is assumed to be **fully charged** at the **beginning** and can be replaced with an **open circuit**.
 2. Find the **time constant** $\tau = RC$.
 - If the circuit has **more than one resistor**, the resistance that we need to find in order to calculate the time constant is the equivalent resistance as seen by the terminals of the capacitor, i.e. the **Thevenin equivalent resistance** $R = R_{Th}$.
 - When possible, this resistance can be obtained by simplification of series or parallel resistances.
 3. Calculate the voltage across the capacitor as $v(t) = V_0 e^{-\frac{t}{\tau}}$.
 4. Find any other circuit variable using the capacitor's voltage.

Note: A **switch** which **opens** or **closes** can **remove** part of the circuit or **add** something to it.

Exercise 3

Find the voltages v_c , v_x , and the current i_x for $t > 0$ if the initial voltage is $v_c(0) = 15$ V.

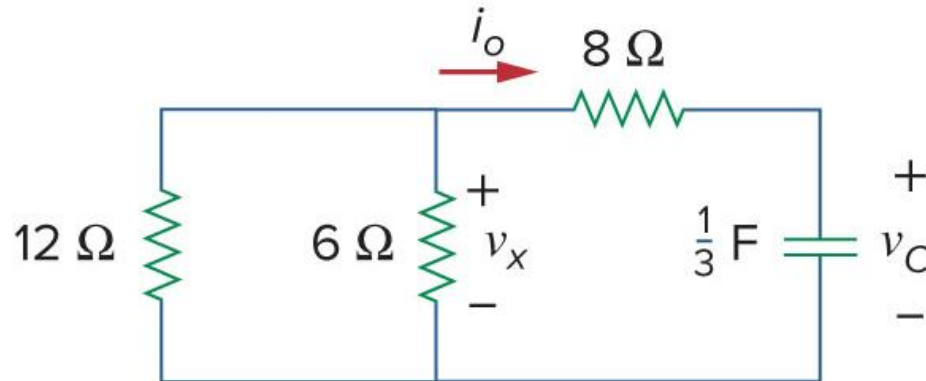
Solution in Exercises_solution_Topic_4.pdf



Exercise 4

Find the voltages v_c , v_x , and the current i_o for $t > 0$ if the initial voltage is $v_c(0) = 60 \text{ V}$.

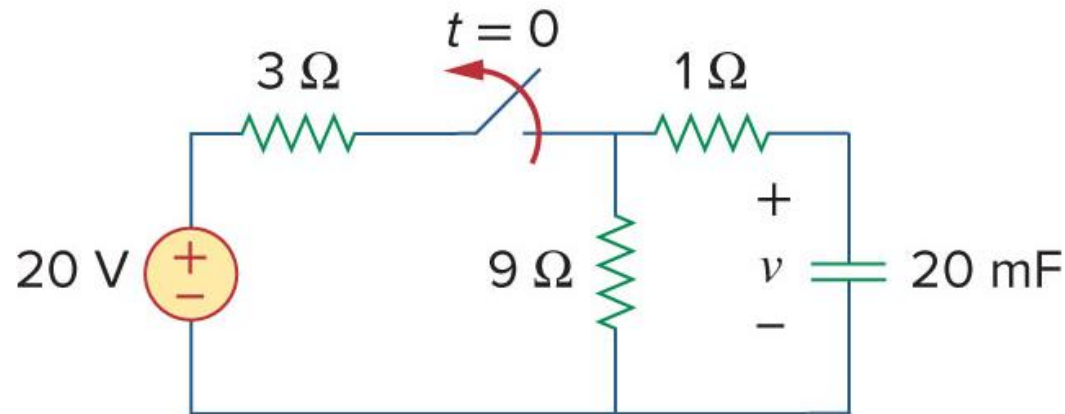
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Exercise 5

The switch in the circuit has been closed for a long time, and it is opened at $t = 0$. Find the voltages $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.

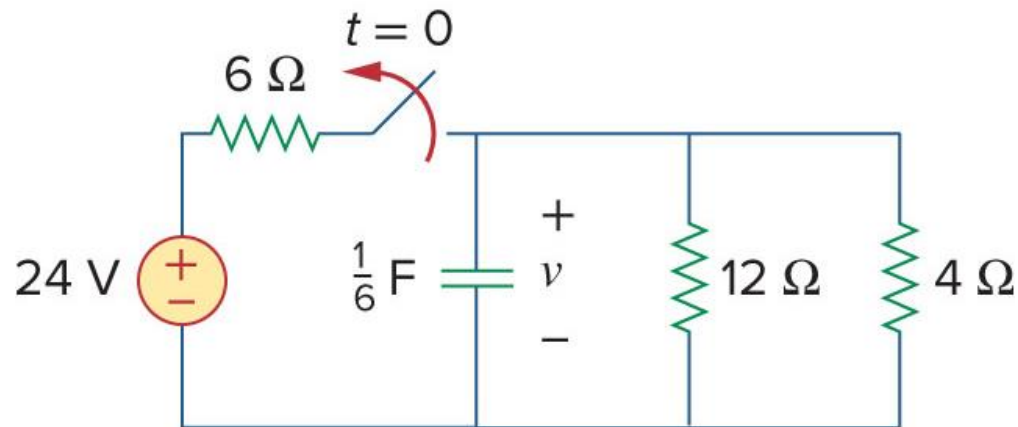
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Exercise 6

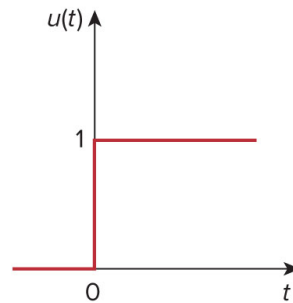
If the switch opens at $t = 0$ after a long time, find the voltages $v(t)$ for $t \geq 0$ and $w_c(0)$.

- For practice!
- Answer: $v(t) = 8e^{-2t}$ V , $w_c(0) = 5.333$ J



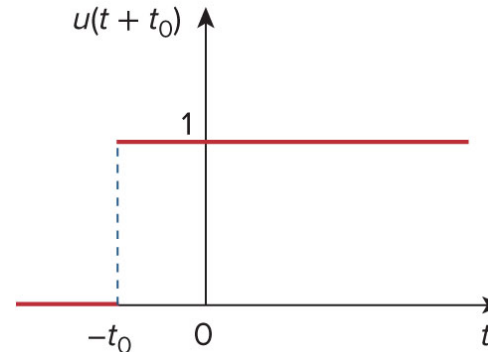
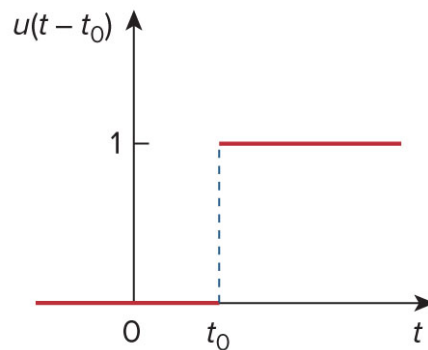
Step response of RC circuit

- A unit step function, denoted by $u(t)$, is zero for negative values of time t and one for positive values of time t (it resembles a step).
 - It serves as a good approximation to switching signals representing a sudden change in voltage or current.



$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

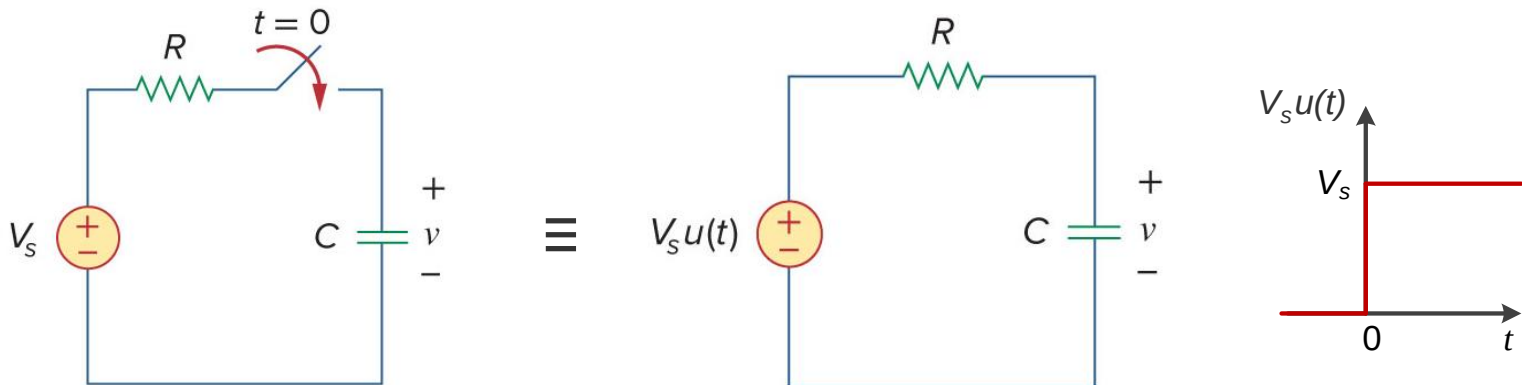
- It can be shifted in time.



Step response of RC circuit

- Step response is the response of the circuit due to a sudden application of a DC voltage or current.
 - It is the circuit behaviour when the **excitation/input** is the **step function**, which may be a voltage or a current source.
 - We can model this behavior with a switch opened or closed at $t = t_0$.
- Let's assume that the capacitor is initially charged to a voltage V_0 . Since the voltage of the capacitor cannot change instantaneously:

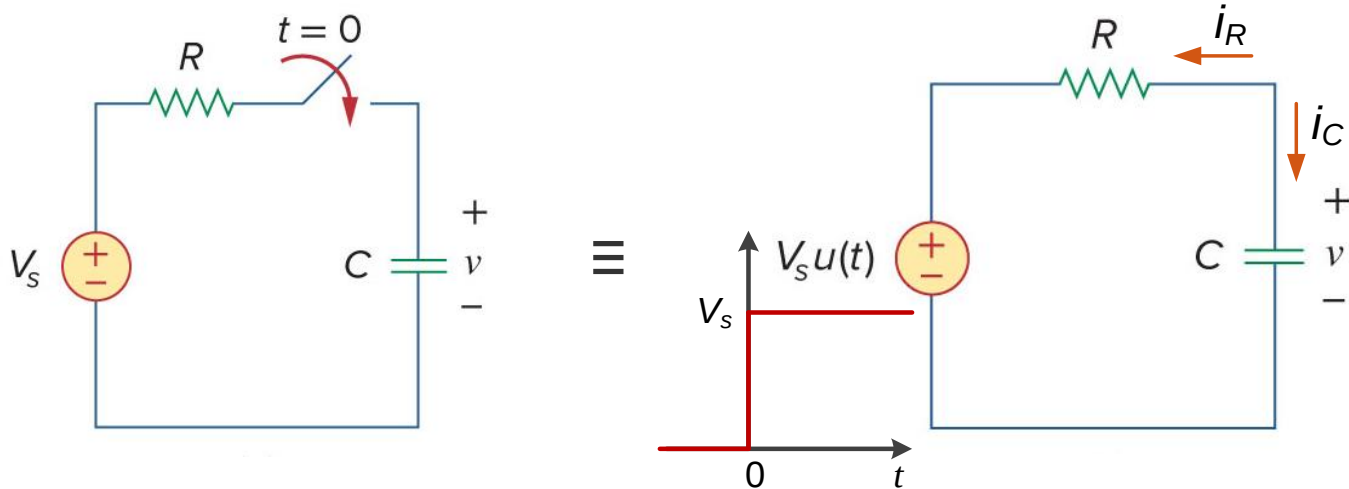
$$v(0^-) = v(0^+) = V_0$$



Note: $v(0^-)$ is the capacitor voltage **just before** switching and $v(0^+)$ **just after** switching.

Step response of RC circuit

To analyse the step response of the RC circuit, we can use KCL at the node between resistor and capacitor.



$$-i_C - i_R = 0 \rightarrow i_C + i_R = 0$$

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

$$C \frac{dv}{dt} + \frac{v - V_s}{R} = 0 \text{ for } t \geq 0$$

Step response of RC circuit

Solve differential equation:

Rearrange:

$$C \frac{dv}{dt} + \frac{v - V_s}{R} = 0, t \geq 0$$

$$\frac{dv}{dt} = \frac{-1}{RC} (v - V_s)$$

Separate:

$$\frac{1}{v - V_s} dv = \frac{-1}{RC} dt$$

Integrate:

$$\int \frac{1}{v - V_s} dv = \frac{-1}{RC} \int dt$$

Solve for $v(t)$:

$$\ln(v - V_s) = \frac{-t}{RC} + D$$

$$v(t) - V_s = e^{\frac{-t}{RC} + D}$$

Apply initial conditions

$$v(0) = V_s + Ae^0 = V_0$$

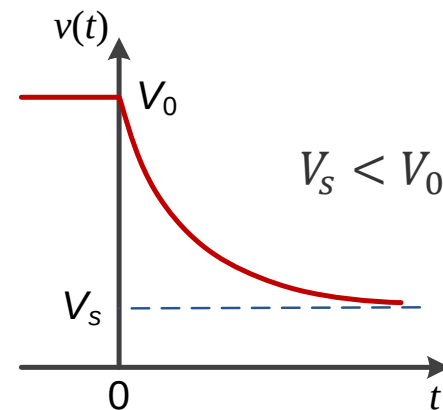
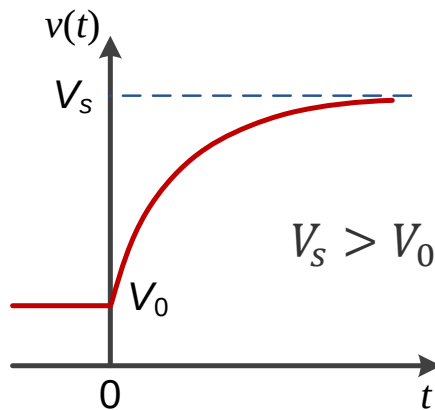
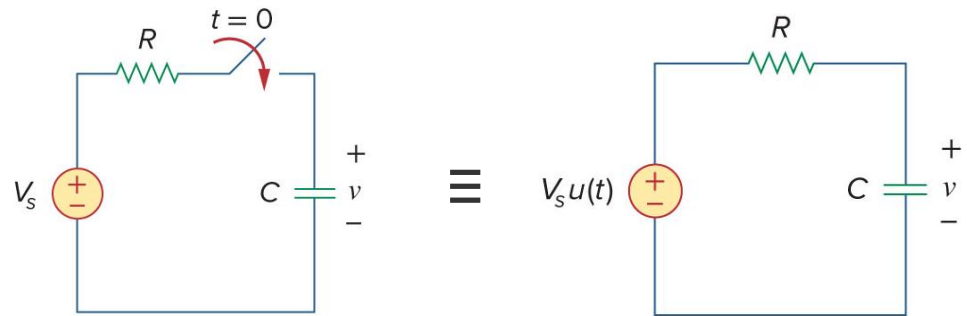
$$v(t) = V_s + Ae^{\frac{-t}{RC}}$$

$$v(t) = V_s + (V_0 - V_s)e^{\frac{-t}{RC}}, t \geq 0$$

Step response of RC circuit

- No need to derive the differential equation solution every time, just use the result.
- The result shows that the **voltage response** of the RC circuit will change from the initial V_0 to the value of V_s in an **exponential manner**.
 - Depending on the value of initial conditions V_0 and voltage source V_s , the capacitor can be charged or discharged.

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-\frac{t}{RC}}, & t > 0 \end{cases}$$

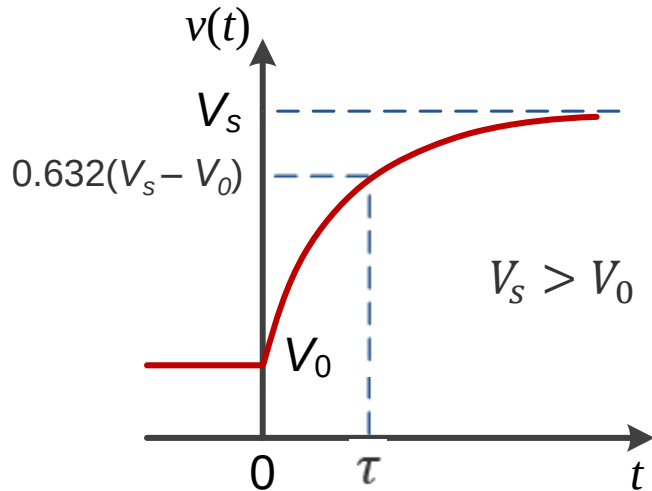


Step response of RC circuit

- The concept of time constant also applies to step response.

The **time constant** of a circuit is the time required for the response to **decay** to $1/e$ (or **36.8%**) of its initial value or to **increase** to $1 - 1/e$ (or **63.2%**) of its final value. It is denoted by τ .

- After **5 time constants** (5τ), the capacitors is **again** considered as **open circuit** as its voltage is no longer changing.



$$\tau = RC$$

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-\frac{t}{\tau}}, & t > 0 \end{cases}$$

Step response of RC circuit

- The step response has **two components**:

$$v(t) = V_s + (V_0 - V_s)e^{-\frac{t}{\tau}}, \quad t > 0$$

$$v(t) = \underbrace{V_0 e^{-\frac{t}{\tau}}}_{\text{Natural response}} + \underbrace{V_s(1 - e^{-\frac{t}{\tau}})}_{\text{Forced response}}, \quad t > 0$$

Natural
response

v_n



Due to stored
energy

Forced
response

v_f



Due to
Independent
sources

Complete response = **natural response** + **forced response**

$$v(t) = v_n(t) + v_f(t), \quad t > 0$$

This can be viewed as **superposition principle** in RC circuits for **two** sources of energy powering the circuit:

1. Initial conditions (stored energy)
2. Independent sources

Step response of RC circuit

- From another perspective:
 - The **transient response** is the circuit's **temporary** response that will **die out** with time.
 - The **steady-state response** is the behaviour of the circuit **a long time after** an external input/excitation is applied (after 5 time constants, 5τ).

$$v(t) = \underbrace{V_s}_{\text{Steady-state response}} + \underbrace{(V_0 - V_s)e^{-\frac{t}{\tau}}}_{\text{Transient response}}, \quad t > 0$$

Steady-state response Transient response

v_{ss}

v_t



Permanent part

Temporary part

Complete response = **transient response** + **steady-state response**

$$v(t) = v_t(t) + v_{ss}(t), \quad t > 0$$

Step response of RC circuit

- More specifically:

1. First stage of steady-state ($t < 0$):

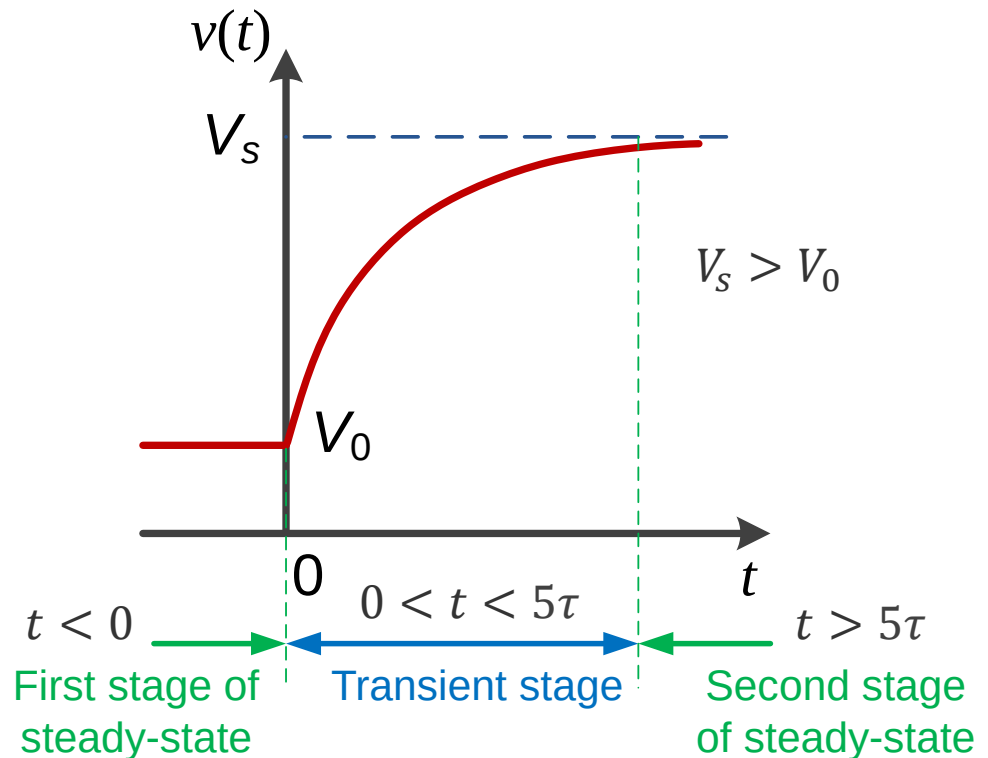
There has been **no change** in the circuit for a **long time** and the capacitor is an **open circuit** with $v(0) = V_0$.

2. Transient stage ($0 < t < 5\tau$):

The capacitor's voltage changes **exponentially**.

3. Second stage of steady-state ($t > 5\tau$):

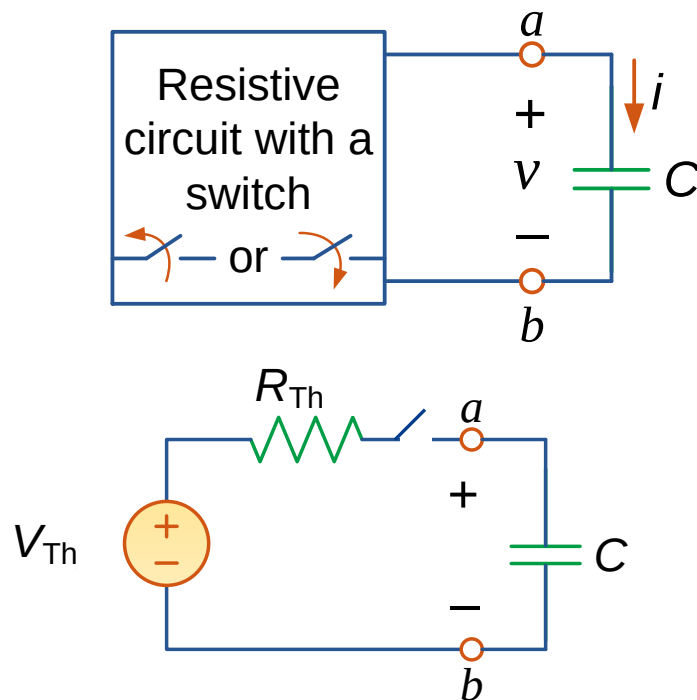
The capacitor's voltage reaches its **final value** or **steady-state value** and becomes **open circuit** again with $v(t) = V_s$ when $t \rightarrow \infty$ or $v(\infty) = V_s$.



$$v(t) = V_s + (V_0 - V_s)e^{-\frac{t}{\tau}}, \quad t > 0$$
$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}, \quad t > 0$$

Step response of (complex) RC circuit

- The step response solution has been derived for a circuit with one single capacitor and resistor.
- It is possible to find the voltage across a capacitor in a complicated circuit with **multiple** resistors, switches, independent and dependent sources by replacing the circuit with its **Thevenin equivalent circuit**.



Step response of RC circuit

- Follow these steps to find step response of RC circuits:
 1. Find the **initial voltage** $v(0)$ at $t = 0$ across the capacitor **before any changes** in the circuit ($t < 0$).
 - The capacitor is assumed to be an **open circuit**.
 2. Find the **final voltage** $v(\infty)$ or V_{Th_∞} at $t \rightarrow \infty$ across the capacitor **after the changes** in the circuit ($t \geq 0$).
 - The capacitor is assumed to be an **open circuit**.
 3. Find the **time constant** $\tau = R_{Th_\infty}C$ **after the changes** in the circuit.
 - R_{Th_∞} is Thevenin equivalent resistance **after the changes** ($t \geq 0$).
 4. Calculate the voltage across the capacitor as:
$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$$
 5. Find any other circuit variable using the capacitor's voltage.

Note: A **switch** which **opens** or **closes** can **remove** part of the circuit or **add** something to it.

Time shift in step response of RC circuit

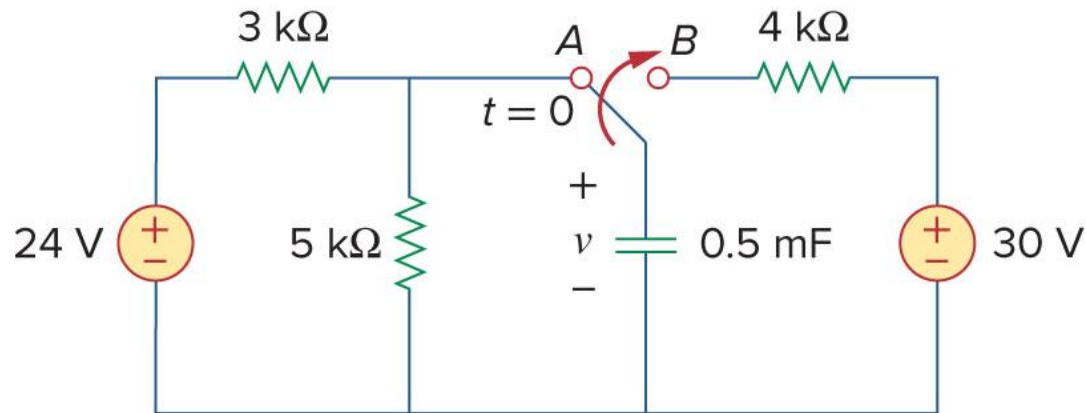
- Note that if the switch changes position at time $t = t_0$ instead of $t = 0$, there is a time delay in the response which can be expressed as time shift in the equation.
 - This method can be used for **multiple switching** at different times.

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{\frac{-(t-t_0)}{\tau}}, \quad t > t_0$$

Exercise 7

The switch in the circuit below has been in position A for a long time. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and $t = 4$ s.

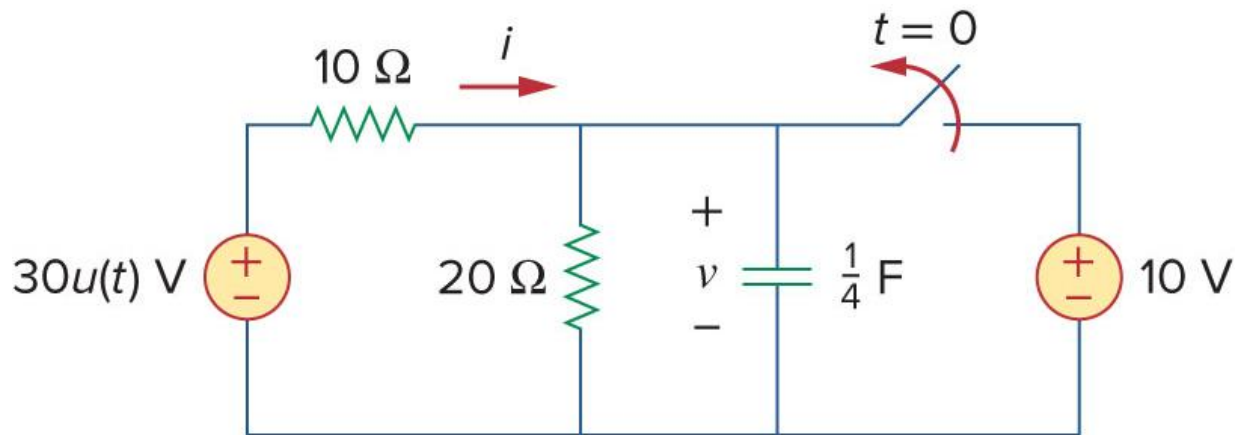
Solution in Exercises_solution_Topic_4.pdf



Exercise 8

In the circuit below, the switch has been closed for a long time and it is opened at $t = 0$. Find v and i for all time.

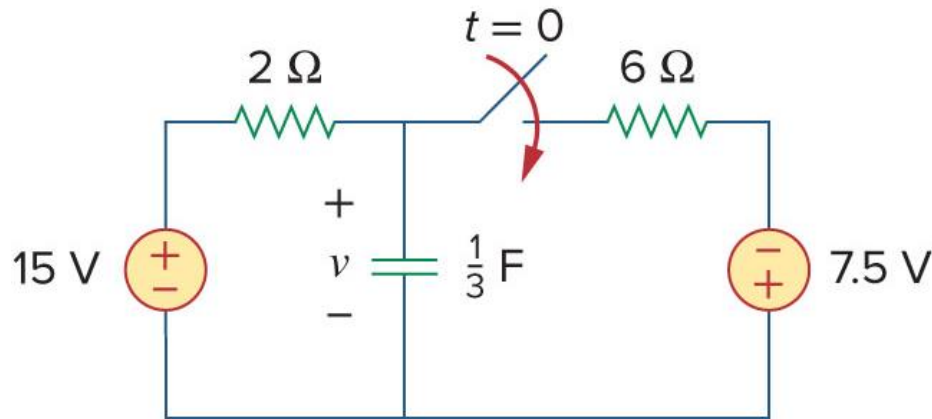
Solution in Exercises_solution_Topic_4.pdf



Exercise 9

Find $v(t)$ for $t > 0$ in the circuit below. Assume the switch has been open for a long time and it is closed at $t = 0$. Calculate $v(t)$ at $t = 0.5$ s.

- For practice!
- Answer: $v(t) = (9.375 + 5.625e^{-2t})$ V, $v(0.5) = 11.444$ V



Questions?

